

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a49314b51fd5d1fb8.jsonl`  
- SHA-256: `fd7705c4082430d4f1ea28ceaf28c53773f44889ccf3635f5f8aceacf60fff87`  
- Source modified: `2026-06-16T03:08:45+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_47057438-9c9`  
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`
```

Transcript

1. user (2026-06-16T03:05:55.396Z)

DESIGN COMPONENT: Threat models + the NON-ATTRIBUTABILITY layer. Enumerate the distinct threat models (public repo browser with NO key; trusted contributor WITH key running tests locally; CI runner; a malicious contributor; a re-identification adversary who runs the same detector on public videos and correlates). For each, state precisely what is achievable and what is NOT (respect the DRM/analog-hole limit: a test-runner can always read what the test reads). Then design the concrete de-linkage / minimization scheme: opaque SALTED fixture IDs (NOT video_id=MD5), stripping of all source metadata (title/url/timestamps/player identity), stream minimization (which feature columns are publishable per the legal memo ? exclude person/pose/biometric and any minors), and the uselessness-in-isolation argument. Tie every choice to the legal memo.

LEGAL MEMO (JSON):

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{  
  "executiveSummary": "INTERNAL MEMO ? general legal information, NOT legal advice; retained India + EU/UK counsel sign-off required before any public commit. QUESTION: can the project commit DERIVED numeric data (per-frame shuttle (x,y) coordinates, rally start/end time-intervals, audio/person/court feature columns, ground-truth intervals) into a public GitHub repo so external contributors can run the rally-segmentation regression suite WITHOUT the videos? BOTTOM LINE: Yes, but only under conditions, and the analysis FORKS on data provenance. The OUTPUT data itself is, across all three jurisdictions, essentially uncopyrightable FACT/measurement ? it does not inherit the source video's copyright (US: Feist + 17 USC 102(b) + NBA v. Motorola; India: EBC v. D.B. Modak, no sui generis right; EU: Football Dataco v. Yahoo ? copyright protects only original selection/arrangement, never the numbers). So a thin compilation copyright at most attaches to a creative schema, never to the coordinates. BUT copyright is the easy axis and NOT where the risk lives. The real exposure is fourfold and provenance- and stream-dependent: (1) the ACT of ingesting/extracting from the source video needs its own lawful basis ? owned/permissioned video is clean; a US fair-use defense for analytic extraction is now CONTESTED, not \"strongly favorable,\" after Thomson Reuters v. Ross (2025) and the US Copyright Office May-2025 report, and unlawful acquisition independently destroys the defense (Bartz v. Anthropic, ~$1.5B settlement); (2) CONTRACT / Terms-of-Service can bar extraction and redistribution even where copyright cannot reach the facts (ProCD; hiQ v. LinkedIn; YouTube ToS bars scraping AND \"harvest information that might identify a person (for example... faces)\" AND redistribution) ? and the remedy is injunction + forced deletion of the committed dataset, not nominal damages; (3) EU/UK sui generis DATABASE RIGHT may subsist in self-derived match-event/coordinate streams because, on the leading Football Dataco v. Sportradar reasoning, recording pre-existing on-court facts you do not control is \"OBTAINING,\" not \"creating\" ? so the project's comfortable \"we created it\" premise is genuinely contestable; (4) PRIVACY ? person/pose/gait streams and anything involving MINORS are the highest-risk and several should NOT be published at all. The cleanest defensible path: commit ONLY owned-or-permissioned-source, minors-excluded, person/pose/gait-EXCLUDED (shuttle + abstract rally-interval + non-biometric feature) data, stripped of every source-attribution field (no filename, no video_id MD5, no match metadata), under a no-extraction-from-third-party-source and no-redistribution hygiene posture, with optional encryption only as a barrier against the no-key public (it cannot bind a contributor who runs the tests).\",
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"derivedDataCoreVerdict": "The derived numeric data DOES NOT inherit the source video's copyright in any of the three jurisdictions ? this is the one firmly-settled conclusion. Per-frame shuttle (x,y) coordinates, rally start/end timestamps, and audio/person/court feature values are uncopyrightable FACTS/measurements about a recorded physical event: discovered, not authored. US: Feist (facts never copyrightable; \"originality, not sweat of the brow, is the touchstone\"), 17 USC 102(b) (no copyright in any \"discovery\"), NBA v. Motorola (facts generated by a sporting event are uncopyrightable \"purely factual information which any patron... could acquire from the arena\" even though the broadcast is protected). A coordinate/timestamp table is also NOT a 17 USC 101 \"derivative work\": a derivative work recasts the original's protected EXPRESSION (framing, editing, imagery, commentary), and a number table carries none of it. India: EBC v. D.B. Modak ? copyright needs \"skill and judgment,\" protects only original arrangement, not facts; no EU-style sui generis right exists. EU: Football Dataco v. Yahoo (C-604/10) ? database copyright protects only an original selection/arrangement; \"the intellectual effort and skill of creating that data are not relevant\" and sweat-of-the-brow cannot ground it. CEILING ON PROTECTION: the MOST that can attach is a THIN compilation copyright in a genuinely creative SELECTION/ARRANGEMENT (e.g., a curated schema) ? never in the raw numbers, and a mechanical/exhaustive layout (every frame, every rally) likely has too little originality even for that (Feist's \"garden-variety listing\" problem). CRITICAL SEPARATION: the STATUS of the output data (settled: unprotected facts) is a different question from the ACT of extraction (transiently copying the video to compute the data), which can independently infringe (Sega v. Accolade: intermediate copying can infringe even when the end product reproduces no protected expression) and which needs its own license or fair-use basis. NON-COPYRIGHT FENCES survive precisely BECAUSE the data is uncopyrightable: enforceable no-extraction ToS (ProCD), narrow hot-news misappropriation (NBA v. Motorola), and the EU/UK sui generis database right. So \"the numbers are free facts\" answers the copyright question and ONLY the copyright question.",

"byJurisdiction": [
 {
 "jurisdiction": "India (PRIMARY ? owner is Bangalore-based, commercialization intended)",
 "copyright": "Derived facts are NOT copyrightable as such: EBC v. D.B. Modak (2008 SC) requires 'skill and judgment' and protects only original arrangement, not facts; India has NO EU-style sui generis database right. BUT this is not as clean as it looks: Indian High Courts protect DATABASES/COMPILATIONS as 'original literary works' under s.2(o) on a low 'skill, labour and judgment' threshold ? Burlington Home Shopping v. Rajnish Chibber, V. Govindan v. E.M. Gopalakrishna Kone, and the OLX v. Padawan order itself (which held a scraped dataset to be a protectable 'proprietary database'). So if numbers are LIFTED from a third party's structured dataset/feed, copying its selection/arrangement can infringe even though the raw facts are free. Self-derived-from-pixels data avoids this; copying an official stats compilation does not.",
 "databaseRight": "No sui generis database right exists in India ? a genuine permissiveness advantage over the EU/UK. Protection runs only through copyright (compilation, above) or contract/IT-Act.",
 "contractToS": "The LIVE risk, not copyright. Breach of anti-scraping ToS plus IT Act s.43 (Government has affirmatively argued public-data scraping 'may be in breach of section 43') and s.66 layering CRIMINAL exposure (up to 3 years / Rs 5 lakh). Enforceability of anti-scraping ToS is untested in Indian courts. OLX v. Padawan was decided AGAINST the scraper; its 'private use' line is obiter, not a safe harbor. Trade-secret/confidentiality for the owner's OWN corpus runs NOT through any US-style DTSA (India has no trade-secret statute) but through the Indian Contract Act 1872, equity, common-law breach of confidence, and IT Act s.72 ? and publication still forfeits the moat under that doctrine.",

"privacy": "DPDP Act 2023 governs personal data of identifiable natural persons; genuinely anonymized numbers fall outside it, but the s.3(c)(ii) 'publicly available data' carve-out is CONTESTED (MeitY/Minister has said scrapers must still meet DPDP consent obligations). MINORS are stricter than the EU: 'child' = under 18 (s.2(f)); s.9(1) requires VERIFIABLE parental consent (DPDP Rules 2025 Rule 10 ? real ID/age verification, not a checkbox); s.9(2) bars detrimental processing; s.9(3) PROHIBITS tracking/behavioural monitoring/targeted ads of children ? subject to narrow Rule 12/Fourth-Schedule class/purpose exemptions that do NOT cover an amateur highlight indexer, so the ban DOES bite this project. A persistent per-player pose/gait signature is, on the conservative reading, prohibited behavioural monitoring; 'tracking' is undefined in the Act, so this is the defensible-conservative, not the certain, reading.",

"netVerdict": "MORE PERMISSIVE than the EU on the database-right axis, but NOT 'lawful on its face.' Defensible to publish ONLY genuinely-anonymized, independently-derived (not lifted from a third-party compilation) numbers, with NO minors tracked and accepting that anti-scraping-ToS + IT Act s.43/s.66 exposure for any third-party-sourced ingestion is unresolved. Owned-source + minors-excluded + no-person/pose/gait = clean."

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    },
    {
      "jurisdiction": "United States (repo + product are global; US fair-use & contract law are the most-developed)",
      "copyright": "Output data does not inherit the video's copyright (Feist; 102(b); NBA v. Motorola); not a 101 derivative work; thin compilation copyright at most. The EXTRACTION act is the open question: fair use for analytic, non-expressive copying was historically favorable (Authors Guild v. Google; Google v. Oracle) but is now CONTESTED ? Thomson Reuters v. Ross (D. Del. 2025) REJECTED the 'copy an expressive work just to reach unprotected facts/data' theory, distinguished the functional-CODE cases (Sega/Oracle), found the use non-transformative, and found Factor-4 harm in a derivative 'data to train AIs' market the plaintiff had not even entered; the US Copyright Office (May 2025) rejected 'non-expressive therefore inherently transformative' and backs a market-dilution harm theory. If the dataset competes with a rights-holder's own data/analytics market, fair use may FAIL. Unlawful source acquisition independently defeats the defense regardless of how transformative the end use is (Bartz v. Anthropic, 2025; ~$1.5B settlement).",
      "databaseRight": "No sui generis database right in the US (Feist rejected sweat-of-the-brow). Competitors stay free to copy raw facts ? subject to the non-copyright fences below.",
      "contractToS": "The dominant US risk. Post-Van Buren the CFAA no longer reaches mere purpose-violations of accessible/public data (though footnote 8 RESERVED whether contract/policy limits can still ground CFAA liability ? the federal-crime route is narrowed, not dead), so platforms enforce via CONTRACT. ToS is an enforceable contract that bites where copyright/CFAA fail (ProCD; hiQ v. LinkedIn ? breach-of-contract summary judgment + $500k consent judgment + injunction + ORDER TO DESTROY scraped-derived data). Binding turns on assent: a logged-in/account/API-key or genuine clickwrap user is bound (hiQ; any YouTube-API user); a purely logged-out scraper under browswrap-only may not be (Meta v. Bright Data) ? and thin 'sign-in wrap' flows can fail for want of conspicuous notice. YouTube ToS verbatim bars automated access, 'harvest any information that might identify a person (for example... faces)', and redistribution/commercial reuse. REDISTRIBUTION (committing the dataset publicly) is the single highest-risk act and the remedy is injunction + forced deletion, not nominal damages.",
      "privacy": "GDPR-style analysis below applies to global users; US state biometric statutes (e.g., Illinois BIPA ? not separately briefed here) and publicity/personality rights are additional fences for face crops and identifiable-player gait. Faces in any committed imagery = identifying personal data; do not publish.",
      "netVerdict": "Output data publishable as facts ONLY if (i) source video was owned/permissioned (or extraction has a defensible fair-use basis you do NOT over-rely on), (ii) no enforceable ToS bars extraction/redistribution of the ingested source, and (iii) no person/pose/gait/face/minor stream is included. Treat third-party-ToS-sourced derived data as NOT publicly committable."
    },
    {
      "jurisdiction": "EU / UK (product/repo global; English-language highlight market makes UK a likely target)",
      "copyright": "Database copyright protects only an original selection/arrangement, never the numbers, and never sweat-of-the-brow (Football Dataco v. Yahoo, C-604/10). Self-derived coordinates are facts, not a copied original structure ? so copyright is a low axis.",
      "databaseRight": "The genuinely contestable axis. The sui generis right protects investment in OBTAINING/verifying/presenting pre-existing data, NOT in CREATING it (BHB v. William Hill C-203/02; Fixtures Marketing) ? which the project reads as 'we create coordinates frame-by-frame, so we are clear.' THAT READING IS OVERCONFIDENT: in Football Dataco v. Sportradar the live match-event data (goals/scorers) was held to be OBTAINED, because the recorder captured pre-existing facts ('events taking place on the pitch') it did NOT control. A badminton rally/shuttle position is the same kind of pre-existing on-court fact recorded rather than authored, so a rights-holder can argue the right SUBSISTS in our streams. Infringement is now governed by CV-Online Latvia (C-762/19, 2021): even a substantial-part taking is actionable only where it adversely affects the maker's ability to recoup investment, balanced against information flow/competition ? a fact-intensive harm test, not a clean 'no protected part = safe' rule. Sportradar (C-173/11) also defeats the server-location dodge for EU-targeted re-utilisation. UK retained its own sui generis right and pre-2021 CJEU case law, so UK-targeted products carry AT LEAST equal, arguably greater, risk; a non-UK/EEA maker may not even hold the right defensively, which does nothing to reduce exposure as a potential infringer.",
      "contractToS": "Ryanair v. PR Aviation (C-30/14): where a database enjoys NEITHER copyright NOR the sui generis right, the Directive's user-protections (incl. no-contracting-out) do not apply, so the source's T&Cs/licence can lawfully BAN automated extraction and redistribution and bind anyone who agreed ? the dominant residual risk. Trap: the licence that

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legitimises INGESTION of the footage is often the very instrument that contractually bans extraction/redistribution of derived data.",

"privacy": "GDPR: shuttle/object coordinates are NOT personal data in isolation (Art.4(1)) ? but coordinates published as part of an identifiable named match are part of a personal-data operation. Person bounding boxes that single out a human are ordinary personal data, NOT Art.9 biometric UNLESS used to uniquely identify (EDPB Guidelines 3/2019). Pose/gait is the danger zone: gait re-identifies even with faces blurred and skeleton 'anonymization' leaves gait identity intact (Weiss et al., Sensors 2025) ? Art.9 special-category the moment used to uniquely identify; explicit-consent basis required. 'Anonymous' is RELATIVE post EDPS v. SRB (CJEU C-413/23 P, 4 Sept 2025): a stream can be non-attributable to a recipient lacking the source yet remains PERSONAL DATA to the holder who can re-run the detector. Minors: Art.8 (consent floor 13?16); any biometric/special-category processing of a child stacks Art.9.",

"netVerdict": "Do NOT assume self-derivation defeats the database right ? treat self-derived event/coordinate streams as potentially 'obtained' data and avoid taking in a way that could reconstitute a rights-holder's dataset or harm their feed's monetisation. Read every source's T&Cs for extraction/redistribution clauses. Keep the footage-copyright analysis (highlight reels) strictly separate. Person/pose/gait/face = do not publish without an Art.9 explicit-consent basis. Get EU + UK jurisdiction-specific advice before redistributing numeric streams at scale."

}

1,

"provenanceFork": "PROVENANCE IS THE DECISIVE FORK ? the legal posture of an identical numeric file flips entirely on where the source video came from. FORK A ? OWNED / OWNER-RECORDED (or fully-permissioned/licensed-for-this-use) SOURCE: This is the clean path and the only one safe to commit publicly. No third-party copyright in the source bars the extraction; no third-party ToS governs the owner's own recording (with two caveats below); the EU/UK database-right 'obtained vs created' debate still applies in theory but there is no adverse rights-holder to assert it against the owner's own footage. The analysis collapses to: (1) a TRADE-SECRET / competitive-moat CHOICE (US framing: 18 USC 1839 DTSA requires secrecy measures + value-from-secrecy; the OPERATIVE framing for this Bangalore owner is Indian breach-of-confidence / Contract Act 1872 / IT Act s.72, NOT the DTSA) ? publishing the data forfeits whatever moat it carried, which is the owner's strategic call to make, weighed against the stated community/benchmark/paper-co-authorship aims; and (2) PRIVACY ? the owner's own footage still contains identifiable people, so DPDP/GDPR and the minors rules below still bind, and person/pose/gait/face streams are still off-limits. TWO CAVEATS that narrow Fork A: (a) 'owner-recorded' is clean ONLY for the owner's OWN non-ticketed/private sessions; footage the owner SHOT AT a third party's event (tournament, club match) carries live exposure ? a ticket is a contract of admission that routinely bars recording/redistribution, and venue/organiser/broadcaster rights (e.g., BWF/league terms, Rome Convention rebroadcast rights) attach independently of any platform ToS; (b) any third-party-collected footage in the corpus (the project's golden set may mix sources) taints those records and pushes them into Fork B. FORK B ? THIRD-PARTY-SOURCED (broadcast / YouTube / scraped / downloaded) SOURCE: Derived data from this provenance SHOULD NOT be committed to a public repo. Even though the numbers are uncopyrightable facts, (1) the EXTRACTION act needs a license or a now-CONTESTED US fair-use defense, and unlawful acquisition (e.g., scraping against ToS, downloading from a platform that bars it) independently sinks the defense (Bartz v. Anthropic); (2) the source's ToS (YouTube et al.) typically bars automated extraction AND redistribution, and committing the derived dataset publicly IS redistribution ? the highest-risk act, remediable by injunction + forced deletion of the committed data (hiQ); (3) re-utilisation/extraction of a substantial part may infringe the EU/UK sui generis database right if the source ties back to a protected feed. PRACTICAL RULE FOR THE REPO: gate the committed corpus to FORK A only. Tag every golden-set record with a verified provenance flag at ingestion; only owned/permissioned-source, non-event-ticketed, minors-excluded, person/pose/gait-excluded records are eligible for the public commit. This maps directly onto the existing split: eval_baselines/ (aggregate JSON snapshots) is already committed precedent and is provenance-agnostic if it carries no per-source attribution; *_golden_features.json are currently kept private (docs/GOLDEN_DATA_SHARING.md) precisely because they derive from a 'Proprietary corpus' ? the right move is to keep Fork-B-tainted features private and publish only a Fork-A, de-attributed subset.",

"privacyVerdict": "PUBLISH/NO-PUBLISH BY STREAM (the project's guardrails already point the right way; this hardens them). PUBLISH-OK (lowest risk): (1) SHUTTLE (x,y) coordinates + Visibility ? describe an inanimate object, NOT personal data on their own (GDPR Art.4(1); WP136 content/purpose/result test) ? PROVIDED they are NOT joined to anything that singles out a named player and NOT published as an identifiable named match (the surrounding personal-data obligations survive even when the coordinate column alone is object-data). (2) Abstract RALLY start/end time-INTERVALS and ground-truth intervals stripped of match/player identity ? facts

about event timing. (3) Non-biometric AUDIO-ONSET, COURT-BOUNDS, and aggregate PERSON COUNT/feature columns used ONLY for rally detection and DETACHED from any identity ? ordinary low-risk data with normal hygiene; NOT Art.9 because the purpose is detect/categorize, not uniquely identify (EDPB Guidelines 3/2019). CONDITIONAL ? only with an Art.9 explicit-consent basis (and effectively never for this v1): PERSON bounding-box coordinates that persistently single out and follow a specific player across rallies ? the purpose flips to unique identification and Art.9 attaches; this architecture is PRONE to that flip, so treat per-player tracking as a tripwire. DO NOT PUBLISH (high risk, exclude from any committed stream): (4) POSE keypoints / SKELETON / GAIT signatures ? gait is a behavioural biometric that re-identifies people even with faces blurred, and pose 'anonymization' provably leaves gait identity intact (Weiss et al., Sensors 2025, 80?95% gait-recognition on anonymized data); raw pose/gait tracks are NOT reliably anonymous, are personal data when usable to single out a player, and become Art.9 special-category once used to uniquely identify ? this matches the project's existing 'gait/biometrics strictly FUTURE (GDPR Art.9)' guardrail. (5) Any FACE CROP or identifiable IMAGERY ? identifying personal data under GDPR and DPDP, and barred by YouTube ToS's face-harvesting clause; do not commit imagery at all. (6) Anything that joins a NAMED PLAYER to a gait/pose signature. MINORS FLIP EVERYTHING: under DPDP a 'child' is anyone under 18; processing ANY child's data needs VERIFIABLE parental consent (Rule 10 ? real ID/age check, not a checkbox), must not cause detrimental effect, and s.9(3) PROHIBITS behavioural monitoring/tracking of children (Rule 12/Fourth-Schedule exemptions do not cover this project, so the ban applies). A persistent per-player pose/gait signature is, conservatively, prohibited monitoring. CLEANEST PATH for any footage containing minors: EXCLUDE minors entirely from any persisted derived stream beyond inanimate shuttle/interval data ? consistent with the existing 'EXCLUDE MINORS from the training pool' guardrail ? or obtain Rule-10 verified parental consent and keep only de-identified object/coordinate data. NON-ATTRIBUTABILITY ARCHITECTURE (the 'secret file' question): Non-attributability is a property of the (data, holder, environment) triple, not of the timeseries. A per-video trajectory/feature stream is effectively a FINGERPRINT (de Montjoye: 4 points re-identify 95% of 1.5M people; k-anonymity collapses on rich quasi-identifiers), so a deterministic detector re-run on a candidate video can correlate-match it to its source ? BUT only if a correlatable reference EXISTS. PUBLIC source: a permanent cheap reference exists, re-linkage is 'reasonably likely', so a raw public-sourced stream should be treated as ATTRIBUTABLE; only heavy aggregation, perturbation/differential privacy, synthetic data, or a non-public enclave (NIST SP 800-188) breaks correlatability. PRIVATE/owner-recorded-and-unpublished source: there is no external reference to correlate against, so source-linkage by an EXTERNAL adversary is not 'reasonably likely' ? BUT it is WRONG to call this 'anonymous by default' (EDPS v. SRB, CJEU C-413/23 P, 4 Sept 2025): to the holder who can re-run the detector, it REMAINS personal data; it is merely RELATIVELY non-attributable to outsiders, and that protection is conditional, revocable (publish or leak the source and the link snaps back), and forward-looking (regulators say assume the data environment grows). HARD ENGINEERING IMPLICATION: the committed fixtures must NOT carry the source filename, match metadata, or the video_id (which is the MD5 of the source file ? using it as a public key would FINGERPRINT/confirm the source the instant anyone holds the file). Use an opaque, random, non-reversible surrogate id per record. Strip the 'name' field in *_golden_features.json. The lead's established constraint is correct: a contributor who RUNS the tests can necessarily read the plaintext fixtures and any key on their machine (DRM/analog-hole limit), so secrecy-from-the-contributor is impossible; the achievable goal is NON-ATTRIBUTABILITY + uselessness-in-isolation, with encryption useful ONLY as a barrier against the no-key public, never against a running contributor.",

"conditionsForSafePublicCommit": [

"PROVENANCE GATE: commit ONLY derived data from owner-recorded / fully-permissioned source video (Fork A). Tag every golden-set record with a verified provenance flag at ingestion; exclude any third-party-sourced (broadcast/YouTube/scraped/downloaded) record from the public commit. For owner-SHOT footage of a third party's event, first clear the admission ticket / venue / organiser / broadcaster terms ? only owner's own non-ticketed sessions are unconditionally clean.",

"SOURCE-LICENCE/ToS CHECK: for every source that touched the committed corpus, confirm in writing that its licence/ToS does NOT bar automated extraction OR redistribution of derived data. Do not rely on US fair use for the extraction step ? it is now contested (Thomson Reuters v. Ross; USCO May-2025) and unlawful acquisition independently defeats it (Bartz v. Anthropic). Redistribution (the public commit) is the single highest-risk act.",

"MINORS EXCLUSION: exclude all footage containing minors (under 18, per DPDP s.2(f)) from any persisted/committed derived stream beyond inanimate shuttle/interval data ? unless Rule-10 verifiable parental consent is held AND only de-identified object/coordinate data is kept. s.9(3)'s behavioural-monitoring ban means a persistent per-player pose/gait signature on a minor is prohibited.",

"BIOMETRIC EXCLUSION: do NOT publish pose keypoints, skeleton tracks, gait signatures, face crops, identifiable imagery, or any named-player-to-movement-signature mapping. Restrict the committed feature columns to shuttle (x,y)+Visibility, abstract rally/GT time-intervals, and non-biometric audio/court/aggregate-person features used only for rally detection and detached from identity.",

"DE-ATTRIBUTION: strip every source-identifying field from the committed fixtures ? source filename, match/event/player metadata, capture dates, and especially the video_id (MD5 of the source file). Replace with an opaque, random, non-reversible surrogate record id. The 'name' field in *_golden_features.json must be removed/aliased.",

"NON-ATTRIBUTABILITY POSTURE: because per-video trajectories are fingerprints correlatable to a source video, achieve non-attributability by keeping the SOURCE private (Fork A, unpublished) ? which makes external source-linkage not 'reasonably likely' ? and/or by coarsening/aggregating/perturbing where feasible. Document the (data, holder, environment) assessment and re-test it as the public-video environment grows (ICO/OVIC forward-looking duty). Treat the streams as personal data IN THE OWNER'S OWN HANDS regardless (EDPS v. SRB).",

"USELESS-IN-ISOLATION + ENCRYPTION-AS-PUBLIC-BARRIER ONLY: keep the committed fixtures useless without the regression harness + schema (abstract intervals + opaque ids, no imagery, no source pointer). Use encryption only as a barrier against the no-key public; do not represent it as protecting against a contributor who runs the tests (they can read plaintext + key locally ? the lead's established DRM/analog-hole constraint).",

"PRECEDENT-CONSISTENT PLACEMENT: publish via the already-committed eval_baselines/ pattern (aggregate, provenance-agnostic JSON snapshots) for headline regression baselines; keep Fork-B-tainted or biometric-bearing *_golden_features.json private (per docs/GOLDEN_DATA_SHARING.md) and publish only a Fork-A, de-attributed, minors-excluded, biometric-free subset of features needed to run the suite.",

"DPDP HYGIENE: ensure any person-derived numeric column that could single out a player is genuinely identity-detached; do not rely on the DPDP s.3(c)(ii) 'publicly available data' carve-out for non-anonymized personal data (it is contested). Maintain purpose-limitation: features persisted for rally detection must not be repurposed for player identification.",

"COUNSEL SIGN-OFF: obtain retained India counsel (DPDP s.9/Rules, IT Act s.43/s.66, breach-of-confidence) AND EU/UK counsel (sui generis database right 'obtained vs created', CV-Online harm test, Ryanair contract) review of the final committed subset and the source-provenance log BEFORE the first public commit. This memo is general information, not advice."

],

"redFlags": [

"OVERCONFIDENCE TRAP #1 ? 'we created the data, so the EU/UK database right cannot apply.' Football Dataco v. Sportradar held live match-event data was OBTAINED (pre-existing on-court facts the recorder did not control), exactly analogous to a recorded badminton rally/shuttle position. A rights-holder can credibly argue the sui generis right subsists in our self-derived streams. Do not bank on the 'created' characterisation.",

"OVERCONFIDENCE TRAP #2 ? 'analytic extraction is strongly favorable fair use (US).' Thomson Reuters v. Ross (2025) REJECTED the 'copy an expressive work to reach unprotected facts' theory and found Factor-4 harm in a derivative data market; the USCO (May 2025) rejected 'non-expressive = inherently transformative.' If the dataset competes with a rights-holder's data/analytics market, fair use may fail. The extraction step is litigable, not settled.",

"OVERCONFIDENCE TRAP #3 ? 'owner's own recordings have no contract problem.' False for footage shot at a third party's event: the admission ticket is a contract barring recording/redistribution, and venue/organiser/broadcaster rights (BWF/league/Rome Convention) attach independently. Only the owner's own non-ticketed sessions are unconditionally clean.",

"OVERCONFIDENCE TRAP #4 ? 'private/unpublished source = anonymous by default.' Per EDPS v. SRB (CJEU 2025) the stream REMAINS personal data to the holder who can re-run the detector; it is only RELATIVELY non-attributable to an external adversary lacking the source, and that is conditional, revocable, and forward-looking. Misclassifying it as 'anonymous' is the exact GDPR error the controller's-perspective holding exists to prevent.",

"FINGERPRINTING / video_id leak: the project's video_id is the MD5 of the source file. Committing it (or any source filename/match metadata) as a public key would let anyone holding the candidate video CONFIRM the source instantly ? defeating non-attributability. Must be stripped and replaced with an opaque surrogate.",

"REDISTRIBUTION REMEDY is not nominal damages: the operative relief in scraping/ToS disputes is INJUNCTION + FORCED DELETION of the derived dataset (hiQ ordered destruction of all scraped-derived code/data). A public commit that breaches a source ToS can be ordered taken down and deleted ? a real, not theoretical, downside.",

"POSE/GAIT mislabeled low-risk: skeleton 'anonymization' leaves gait identity intact (80?95% re-ID), so any pose/gait stream is NOT reliably anonymous and is Art.9-adjacent. Excluding it is not optional polish ? it is a guardrail.",

"MINORS + behavioural monitoring: DPDP s.9(3) bars behavioural monitoring of under-18s and the Rule 12/Fourth-Schedule exemptions do not cover this project. A persistent per-player pose/gait signature on a minor is, conservatively, a prohibited act with criminal-adjacent regulatory exposure ? exclude minors from any persisted stream beyond inanimate shuttle/interval data.",

"MIXED-PROVENANCE corpus: the golden set may blend owned and third-party footage. A single Fork-B-tainted record contaminates the public commit. Provenance must be verified per-record, not assumed corpus-wide.",

"WRONG-LAW trade-secret framing: analysing the owner's moat under US 18 USC 1839 (DTSA) is a jurisdictional error ? the Bangalore owner's protection runs through Indian breach-of-confidence / Contract Act 1872 / IT Act s.72. The publish-forfeits-the-moat intuition survives, but the operative doctrine is different."

],

"counselQuestions": [

"EU/UK sui generis database right: on the Football Dataco v. Sportradar 'obtained vs created' reasoning, does the right subsist in our self-derived shuttle-coordinate / rally-event streams, and if so does redistributing a de-attributed regression subset 'extract/re-utilise a substantial part' and cause CV-Online-Latvia-type harm to any rights-holder's feed? (EU AND UK, given likely English-language market.)",

"US fair use for the EXTRACTION step, post-Thomson Reuters v. Ross and the USCO May-2025 report: is transient analytic copying of a source video to compute rally/coordinate features defensible where the resulting dataset could be seen to compete in a 'data/analytics' market ? and does it matter that we only ever publish numbers, never the footage?",

"India: are anti-scraping/ToS terms enforceable in Indian courts, and what is the realistic IT Act s.43 (civil) / s.66 (criminal) exposure for ingesting third-party footage to derive features ? and does the pending ANI v. OpenAI judgment (reserved, ~April 2026) change the fair-dealing/TDM analysis?",

"India DPDP s.9 + Rules 10/12: for a corpus that may contain players under 18, what verifiable-parental-consent and exemption analysis applies, and does a persistent per-player pose/gait (or even per-player rally-tracking) feature constitute prohibited 'behavioural monitoring/tracking' of a child under s.9(3)?",

"Owner-shot event footage: do the admission tickets / venue / organiser (e.g., BWF, local league) / broadcaster terms for the specific tournaments and clubs in our corpus bar recording, derivation, or redistribution of derived data ? and which records must therefore be excluded from the public commit?",

"Personal-data classification under GDPR + DPDP for the SPECIFIC committed feature columns: at what point do identity-detached person/court/coordinate features 'single out' a player (especially in a small known squad/roster), and does our de-attribution + opaque-surrogate-id scheme suffice to keep the published subset outside personal-data obligations for downstream recipients (per EDPS v. SRB)?",

"Trade-secret / breach-of-confidence (Indian law): given commercialization and possible paper co-authorship, what is the moat trade-off of publishing the Fork-A subset, and are there confidentiality/Contract-Act 1872 / IT Act s.72 steps to take before forfeiting secrecy on any owner-recorded derived data?",

"Source-licence audit: for each source category in the corpus (owner-recorded, permissioned partner, downloaded/scraped), confirm the licence/ToS expressly permits BOTH automated extraction AND public redistribution of derived numeric data ? and where silent or prohibitive, which records are ineligible.",

"Encryption + 'secret file' design: does adding encryption that a running contributor can necessarily decrypt create any misrepresentation/false-security or DPDP 'reasonable security safeguards' compliance question, and is non-attributability-via-private-source a defensible posture to assert to regulators given its conditional/revocable nature?",

"Cross-border data transfer: does committing any record that is (or may be argued to be) personal data to a global public GitHub repo implicate DPDP cross-border-transfer rules or GDPR international-transfer obligations, independent of the copyright/contract analysis?"

],

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judgment-ai-copyright-india/"

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    }
  ]
}

```

GROUNDING:

CODEBASE FACTS (verified by the lead before this run):

- eval_baselines/ is a TRACKED (committed) dir of JSON baseline snapshots (regression_baseline.json, heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json, experiment_leaderboard.json). Precedent: committed structured regression files already live in git.
- backend/eval/regression.py loads a committed baseline, scores DB-stored predictions vs ground-truth intervals, flags P/R drop beyond tolerance, with incommensurability guards

(IoU/detector/GT-hash). Predictions come from the DB (needs a pipeline run on the video) -> NOT video-free on the prediction side yet.

- backend/eval/fusion_golden.py extracts *_golden_features.json from VIDEO+trajectory CSV+labels (needs GPU/WSL). backend/eval/fusion_compare.py & backend/eval/ablation.py RE-SCORE those JSON features on pure CPU with NO video (LOVO deltaF1, bootstrap CI, sign test). This is the EXISTING VIDEO-FREE seam.

- docs/GOLDEN_DATA_SHARING.md (#175): *_golden_features.json are deliberately kept PRIVATE on OneDrive, OUT of git, because they derive from a 'Proprietary corpus.' Existing skipif litmus tests (tests/test_ablation.py::test_litmus_reproduces_gate_a, tests/test_fusion_compare.py) run ONLY if those features are present in output/ -> currently skipped everywhere.

- Data contracts: trajectory CSV columns (Frame,Visibility,X,Y); candidate/final rally segments = (start,end) time intervals; multi-signal fusion uses shuttle + person + audio + court-bounds feature columns (weak signals retained, default-OFF). video_id = MD5 of the source file (backend/utils/hashing.py) ? i.e. using it as a public key would FINGERPRINT the source file.

- Guardrails: commercial-clean licensing MIT/Apache/BSD only; paid LLMs never a training source; EXCLUDE MINORS from the training pool; gait/biometrics strictly FUTURE (GDPR Art 9). Owner is Bangalore/India-based; aims at commercialization and possibly co-authoring a research paper.

- A SEPARATE workflow is concurrently designing the general video-free fixture mechanics. THIS effort focuses on (a) the LEGAL/IP/privacy status of committing DERIVED structured data PUBLICLY, and (b) a non-attributability + encryption ('secret file') architecture.

GOAL: let an external contributor on another clone run the rally-segmentation regression suite from COMMITTED structured fixtures (no video present), such that the fixtures (i) cannot be attributed to a specific source video/match/person and (ii) are useless to a public user in isolation.

KEY CONSTRAINT ALREADY ESTABLISHED BY THE LEAD: a contributor who RUNS the tests necessarily can read the plaintext fixtures + any key on their machine (DRM/analog-hole limit). So secrecy-from-contributor is impossible; the achievable goal is NON-ATTRIBUTABILITY + uselessness-in-isolation, plus encryption only as defense against the NO-KEY public.

2. assistant (2026-06-16T03:05:59.349Z)

I'll analyze the codebase facts and the legal memo to design the threat models and non-attributability layer. Let me first verify the key codebase facts referenced in the grounding.

3. user (2026-06-16T03:06:02.377Z)

Already up to date.

4. user (2026-06-16T03:06:02.534Z)

```
experiment_leaderboard.json
gemini_wrapper_2026-06-10.json
heuristic_lovo_n6.json
---DOCS---
docs/GOLDEN_DATA_SHARING.md
```

5. assistant (2026-06-16T03:06:07.092Z)

Let me verify the specific code seams referenced ? the golden features schema, the eval/fusion modules, and the video_id helper.

6. user (2026-06-16T03:06:08.605Z)

```
1      # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3      **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4      **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts
(trajecory CSVs, `*_golden_features.json`, manifests, baseline snapshots) so the **GPU box**
and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6      ## Why this exists
7      The rally-quality measurement is split across two machines by capability:
8      - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **.rallies.csv` golden labels**, and a **GPU/WSL**
```

```

stack ? only the GPU box can run it.
9      - ``backend/eval/fusion_compare.py`` / ``backend/eval/ablation.py`` (the LOVO
``litmus``: ?F1, bootstrap CI, paired sign test) are ``pure-CPU re-scoring`` that read only the
small ``*_golden_features.json`` ? any session (incl. a CPU dev container) can run them.
10
11      Today those artifacts sit in gitignored ``output/`` (per-machine) and scattered local
folders (``...\Annotation Setup\Collect\Trajectories\``, ``...\Golden Labelled\``). Nothing syncs
them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap
with OneDrive.
12
13      ## Canonical shared folder
14      ``%USERPROFILE%\OneDrive\rally-annotator\golden-data\`` ?
``C:\Users\avidu\OneDrive\rally-annotator\golden-data\``
15
16      OneDrive is present + signed-in on both the GPU box and the dev machine under the same
account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an
overwrite):
17
18      ```
19      golden-data/
20      README.md
21      features/<corpus-tag>/      *_golden_features.json      (small ? the key shared
artifact)
22      trajectories/<corpus-tag>/  *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23      manifests/                  *_lovo_manifest.json
24      baselines/                  leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25      ```
26
27      ``<corpus-tag>`` = e.g. ``2026-06-15-n6`` (date + corpus size). ``Do NOT`` put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29      ## The GPU ? CPU workflow
30      1. ``GPU box`` (videos + CSVs + labels present):
31      ```
32      python -m backend.eval.fusion_golden --manifest <manifest> \
33      --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"
34      ```
35      (or extract to local ``output/`` then copy the ``*_golden_features.json`` into the shared
folder).
36      2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
37      3. ``CPU dev/agent session`` points the analysis at the shared folder:
38      ```
39      python -m backend.eval.fusion_compare --features-dir "...golden-
data\features\2026-06-15-n6" --record
40      python -m backend.eval.ablation      --features-dir "...golden-
data\features\2026-06-15-n6" --granularity group
41      ```
42      ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's ``output/``
to also light up the ``skipif`` golden-litmus tests
(``tests/test_ablation.py::test_litmus_reproduces_gate_a``, ``tests/test_fusion_compare.py``).
43
44      ## Optional code ergonomics (in #175, not required)
45      - A ``RALLY_GOLDEN_DIR`` env var (default ``output/``) honored by ``fusion_golden`` /
``fusion_compare`` / ``ablation`` so neither machine hardcodes the path. (The CLI ``--out-dir`` /
``--features-dir`` flags already make this a convention; the env var is just ergonomics.)
46      - A tiny ``scripts/sync_golden.py`` to copy ``output/*_golden_features.json`` ? the shared
folder.
47
48      ## Guardrails
49      - The shared folder lives ``outside the repo`` ? never committed. Keep ``output/``
gitignored and do ``not`` symlink the shared folder into a tracked path.
50      - The features derive from the ``Proprietary`` corpus ? keep the OneDrive folder

```

****private**** (no public share links).

51

52 See also: `docs/NEXT\_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (P3), `docs/TRACKNET\_WSL\_SETUP.md` (the GPU/WSL detector
setup).

53

7. user (2026-06-16T03:06:09.030Z)

__init__.py
__pycache__
ablation.py
annotations.py
audio
batch.py
calibrate_local.py
calibrate_wasb.py
calibration.py
classifier.py
distill_local.py
eval_partial_labels.py
eval_stats.py
experiment.py
export_wasb_segments.py
fusion_audio.py
fusion_compare.py
fusion_golden.py
gemini_refine.py
gt_loader.py
harness.py
metrics.py
per_user_eval.py
per_user_gate.py
per_user_regression.py
promotion.py
rally_seq_proto.py
regression.py
score_calibration.py
segmentation_metrics.py
serve_contrast.py
served_gate_a.py
splits.py
training.py
windowing.py

8. assistant (2026-06-16T03:06:12.919Z)

Now let me examine the golden features schema and the video_id helper to ground the de-attribution scheme in the actual data contracts.

9. user (2026-06-16T03:06:13.415Z)

1 """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3 Mirrors `distill\_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4 candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5 EXISTING golden trajectory CSVs (so ****no WASB re-run****), needs ****no Gemini and no DB****,
and
6 writes one small JSON per video. The LOVO comparison (`fusion\_compare.py`) then runs
100% on
7 CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8 measured by the shipped experiment harness; person features enter only as feature
columns the

```

9      learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11      Run (on the GPU box):
12      python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14      Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15      `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16      """
17      from __future__ import annotations
18
19      import argparse
20      import json
21      import os
22      import time
23      from typing import Any, Dict, List, Optional
24
25      from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
26      from backend.pipeline.detectors import rally_gate
27      from backend.pipeline.detectors import fusion_features as ff
28      from backend.eval import distill_local
29      from backend.eval.calibrate_wasb import load_golden_gts
30      from backend.eval.training import label_candidates
31      from backend.utils.run_telemetry import RunTelemetry
32
33      # The 5 fps/resolution-invariant shuttle features distill_local already computes.
34      SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
35      # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
36      PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
37
38
39      def _derive_video_path(spec: Dict[str, Any]) -> str:
40          """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
41          golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
convention)."""
42          if spec.get("video_path"):
43              return str(spec["video_path"])
44          labels = str(spec["labels"])
45          if labels.endswith(".rallies.csv"):
46              return labels[: -len(".rallies.csv")] + ".MP4"
47          raise ValueError(f"{spec.get('name')}: no video_path and labels {labels!r} isn't
*.rallies.csv")
48
49
50      def extract_video(spec: Dict[str, Any], detector: Optional[Any] = None,
51                      frame_skip: int = 3, iou: float = 0.5,
52                      log_every_sec: float = 15.0,
53                      telemetry: Optional[Any] = None) -> Dict[str, Any]:
54          """One golden video -> a feature record (shuttle + person features per candidate
window).
55
56          Returns `{name, fps, frame_width,
candidates:[{start,end,shuttle{}},person{},label]}, gts}`.
57          `detector` (anything with `detections_over_windows`) is injectable for GPU-free
tests;
58          when None a real `PersonDetector` is built (loads torchvision on first use).
59          `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health
snapshots
60          from the detection progress callback so a hard kill leaves a forensic runlog.
61          """
62          fps, fw = float(spec["fps"]), float(spec["frame_width"])
63          traj = str(spec["trajectory"])

```



```

64     points = TrackNetRunner.parse_trajectory_csv(traj)
65     intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
66     gts = load_golden_gts(str(spec["labels"]))
67     labels = label_candidates(intervals, gts, iou)
68     # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
estimate
69     # (parity with the shuttle net-crossing gate) rather than hard-coding an
orientation.
70     axis, net_px, _span = rally_gate.estimate_play_axis(points)
71     video_path = _derive_video_path(spec)
72
73     if detector is None:
74         from backend.pipeline.detectors.person_detector import PersonDetector
75         detector = PersonDetector({"indexing": {"use_gpu": True}})
76
77     # Single-pass person detection over ALL candidate windows (open the video ONCE, read
78     # forward ? no per-window seek). Then compute features per window (pure, no GPU).
79     n = len(intervals)
80     win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]
81     print(f"[fusion_golden]   {spec['name']}: {n} windows ? person-detecting in one
pass...", flush=True)
82     t0 = time.monotonic()
83     state = {"last": t0}
84
85     def _progress(done: int, total: int) -> None:
86         if telemetry is not None:
87             telemetry.snapshot(done, total, phase=spec["name"])
88             now = time.monotonic()
89             if now - state["last"] >= log_every_sec or done >= total:
90                 el = now - t0
91                 rate = done / el if el > 0 else 0.0
92                 eta = (total - done) / rate if rate > 0 else 0.0
93                 pct = 100 * done // total if total else 0
94                 print(f"[fusion_golden]   {spec['name']}: detect frame {done}/{total}
({pct}%) "
95                       f"| {el:.0f}s elapsed | ETA {eta:.0f}s", flush=True)
96                 state["last"] = now
97
98     per_window = detector.detections_over_windows(video_path, win_frames, frame_skip,
99                                                    progress_cb=_progress)
100
101     candidates: List[Dict[str, Any]] = []
102     for i, ((s, e), xs) in enumerate(zip(intervals, x_shuttle)):
103         det = per_window[i]
104         det_fw = det.get("frame_width") or fw
105         det_fh = det.get("frame_height") or 0.0
106         # Normalise the (pixel) net coord to 0-1 on the play axis, matching the foot-
point
107         # normalisation fusion_features uses (x/width, y/height).
108         if axis == "x":
109             net_norm = net_px / det_fw if det_fw > 0 else 0.5
110         else:
111             net_norm = net_px / det_fh if det_fh > 0 else 0.5
112         sf, ef = win_frames[i]
113         shuttle_in = [p for p in points if sf <= p.frame <= ef]
114         person = ff.compute_person_features(det["frames"], shuttle_in, det_fw, det_fh,
115                                             court_polygon=None, net=(axis, net_norm))
116         candidates.append({
117             "start": float(s), "end": float(e),
118             "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
119             "person": {k: float(v) for k, v in person.items()},
120             "label": int(labels[i]),
121         })
122     return {
123         "name": spec["name"], "fps": fps, "frame_width": fw,

```

```

124         "candidates": candidates,
125         "gts": [[float(a), float(b)] for a, b in gts],
126     }
127
128
129     def run(manifest_path: str, out_dir: str, frame_skip: int = 3, iou: float = 0.5,
130            fresh: bool = False, smallest_first: bool = False) -> List[str]:
131         with open(manifest_path, encoding="utf-8") as f:
132             specs = json.load(f)
133         if smallest_first:
134             # Process small videos first for quick feedback (the slow full match lands
135             # last);
136             # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
137             # row/frame).
138             specs = sorted(specs, key=lambda s: (os.path.getsize(s["trajectory"])
139             if os.path.isfile(s.get("trajectory", ""))
140             else 0))
141         os.makedirs(out_dir, exist_ok=True)
142         # Run-telemetry black box: a long GPU run can be OS-killed with no traceback, so
143         # keep a tiny
144         # rotating runlog of velocity + machine health for the post-mortem. Constructing it
145         # must never
146         # break extraction, so guard the build and fall back to no telemetry on any error.
147         telem: Optional[RunTelemetry]
148         try:
149             telem = RunTelemetry(os.path.join(out_dir, "run-telemetry.jsonl"),
150             label="fusion_golden")
151         except Exception: # noqa: BLE001 - telemetry wiring must never break the run it
152             # observes
153             telem = None
154         written: List[str] = []
155         for vi, spec in enumerate(specs, 1):
156             out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
157             # Resumable: skip a video whose JSON already exists (a re-run picks up where a
158             # prior
159             # run left off instead of re-detecting). --fresh forces a redo.
160             if not fresh and os.path.isfile(out):
161                 print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already
162                 extracted "
163                 f"-> {out} (use --fresh to redo)", flush=True)
164                 written.append(out)
165                 continue
166             print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
167             flush=True)
168             t0 = time.monotonic()
169             if telem is not None:
170                 telem.event("video_start", spec["name"])
171             try:
172                 data = extract_video(spec, frame_skip=frame_skip, iou=iou, telemetry=telem)
173             except Exception as e: # noqa: BLE001 - record machine state at death, then re-
174                 # raise
175                 if telem is not None:
176                     telem.event("error", spec["name"], err=str(e))
177                 raise
178             n_pos = sum(c["label"] for c in data["candidates"])
179             with open(out, "w", encoding="utf-8") as f:
180                 json.dump(data, f, indent=2)
181             if telem is not None:
182                 telem.event("video_done", spec["name"],
183                 windows=len(data["candidates"]), positives=n_pos)
184             print(f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: DONE in
185             {time.monotonic() - t0:.0f}s "
186             f"| {len(data['candidates'])} windows ({n_pos} positive) |
187             {len(data['gts'])} golden rallies "
188             f"-> {out}", flush=True)

```

```

176         written.append(out)
177     return written
178
179
180     def main() -> None:
181         ap = argparse.ArgumentParser(description="Extract golden-set shuttle+person features
(fusion P3 GPU step).")
182         ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
183         ap.add_argument("--out-dir", default="output")
184         ap.add_argument("--frame-skip", type=int, default=3)
185         ap.add_argument("--iou", type=float, default=0.5)
186         ap.add_argument("--fresh", action="store_true", help="re-extract even if a JSON
already exists")
187         ap.add_argument("--smallest-first", action="store_true",
188             help="process smaller videos first for quick feedback (LOVO is
order-independent)")
189         args = ap.parse_args()
190         run(args.manifest, args.out_dir, args.frame_skip, args.iou, fresh=args.fresh,
191             smallest_first=args.smallest_first)
192
193
194     if __name__ == "__main__":
195         main()
196

```

10. user (2026-06-16T03:06:13.971Z)

```

1     """Canonical file-identity hashing for the indexer.
2
3     `video_id` is the MD5 of the *whole* video file. Historically each entriypoint
4     (`backend/main.py`, `run_eval.py`, `scripts/run_phase3_eval.py`,
5     `scripts/run_process_and_stitch.py`)
6     defined its own `get_file_md5` with a different read-chunk size (64KB / 8KB / 1MB).
7     The hash values agreed (same bytes hashed), but the duplicated code was a footgun:
8     any future change to one copy would silently diverge the DB key.
9
10    This module is the single source of truth. Import `compute_video_id` everywhere a
11    video needs to be keyed.
12    """
13
14    from __future__ import annotations
15
16    import hashlib
17
18    # 64 KB matches the value used by the live pipeline (backend/main.py) before this
19    # was centralized. The chunk size does not affect the resulting digest ? it only
20    # bounds memory use while streaming large files.
21    _CHUNK_SIZE = 65536
22
23    def compute_video_id(filepath: str) -> str:
24        """Return the MD5 hex digest of the whole file ? the canonical video_id.
25
26        Streams the file in fixed-size chunks so multi-gigabyte recordings never load
27        into memory at once.
28        """
29        hasher = hashlib.md5()
30        with open(filepath, "rb") as f:
31            for chunk in iter(lambda: f.read(_CHUNK_SIZE), b""):
32                hasher.update(chunk)
33        return hasher.hexdigest()
34

```

11. assistant (2026-06-16T03:06:19.894Z)

I now have the exact data contracts. The golden features JSON carries `name` (the source-derived video name, per `\_derive\_video\_path` ? directly tied to the filename and `.rallies.csv` golden labels), `fps`, `frame\_width`, per-candidate `shuttle`/`person` feature dicts, `label`, and absolute `start`/`end`/`gts` time intervals. The `person` block is the biometric tripwire. The `name` and the MD5 `video\_id` are the fingerprinting vectors.

Let me confirm the person feature names (to know exactly which columns the legal memo flags as conditional/excluded) and check the ablation/fusion_compare schema expectations so the de-attribution scheme stays runnable.

12. user (2026-06-16T03:06:21.938Z)

```
194:PERSON_FEATURE_NAMES = [
---
41:FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
```

13. user (2026-06-16T03:06:23.368Z)

```
37:     feature columns + golden gts). The harness derives labels from gts itself.""
42:     cand = d["candidates"]
46:         features[fn] = [float(c["shuttle"][fn]) for c in cand]
48:         features[fn] = [float(c["person"][fn]) for c in cand]
49:     gts = [(float(a), float(b)) for a, b in d["gts"]]
50:     videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
51:                                             features=features, gts=gts))
```

14. user (2026-06-16T03:06:26.754Z)

```
194     PERSON_FEATURE_NAMES = [
195         "players_in_court", "two_sided_motion", "mean_player_motion",
196         "in_court_ratio", "shuttle_player_colocation",
197     ]
198
199
200     def compute_person_features(per_frame: Dict[int, List[Tuple[int, BBox]]],
shuttle_points,
201                                     frame_width: float, frame_height: float,
202                                     court_polygon: Optional[Sequence[Tuple[float, float]]] =
None,
203                                     net: Optional[Net] = None,
204                                     min_players: int = 2) -> Dict[str, float]:
205         """Aggregate the per-window person features from per-frame detections + the in-
window
206         shuttle trajectory. Pure; returns exactly `PERSON_FEATURE_NAMES`, all floats."""
207         counts = in_court_counts(per_frame, frame_width, frame_height, court_polygon)
208         return {
209             "players_in_court": players_in_court(counts),
```

15. user (2026-06-16T03:06:27.602Z)

```
30     by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
31     vids = [p["video"] for p in res["variant_a"]["per_video"]]
32     return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])
33
34
35     def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
36         """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
37         feature columns + golden gts). The harness derives labels from gts itself."""
38         videos: List[experiment.VideoCandidates] = []
39         for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
40             with open(path, encoding="utf-8") as f:
41                 d = json.load(f)
42                 cand = d["candidates"]
```

```

43         intervals = [(float(c["start"]), float(c["end"])) for c in cand]
44         features: Dict[str, List[float]] = {}
45         for fn in SHUTTLE_FEATURE_NAMES:
46             features[fn] = [float(c["shuttle"][fn]) for c in cand]
47         for fn in PERSON_FEATURE_NAMES:
48             features[fn] = [float(c["person"][fn]) for c in cand]
49         gts = [(float(a), float(b)) for a, b in d["gts"]]
50         videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
51                                                    features=features, gts=gts))
52     return videos
53
54

```

16. assistant (2026-06-16T03:06:37.928Z)

I now have everything I need. The schema is fully grounded:

- The committed fixture's `name` field flows straight into `VideoCandidates(name=d["name"], ...)` ? it's the source-derived video name (from the `.MP4`/`.rallies.csv` filename), a direct attribution vector. The loader uses it only as a label, so it can be aliased to an opaque surrogate without breaking the harness.
- `intervals = [(start, end)]` and `gts = [[a,b]]` are **absolute** time intervals (seconds into the source video) ? these tie back to the match timeline. The harness only needs them **relative** to each other (IoU between candidate and GT), so a per-fixture time-origin shift preserves all scoring math while destroying timeline-correlatability.
- The `person` block (`players\_in\_court`, `two\_sided\_motion`, `mean\_player\_motion`, `in\_court\_ratio`, `shuttle\_player\_colocation`) ? aggregate per-window person features ? is exactly the stream the legal memo flags as the Art.9 tripwire ("this architecture is PRONE to that flip"). These are **aggregate counts/motion**, not pose/gait/skeleton, so they are NOT in the "do-not-publish" pose/gait bucket ? but they are the conditional bucket, and a small-known-roster singling-out risk applies.
- The shuttle block (`duration`, `peak\_velocity`, `mean\_velocity`, `active\_ratio`, `net\_crossings`) is fps/resolution-invariant object data ? the publish-OK stream.

I have what I need to write the design. Let me confirm the experiment harness reads `gts`/`intervals` only relatively (IoU), so the time-origin shift is safe.

17. user (2026-06-16T03:06:39.570Z)

```

4:tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5:start < end. We score a list of *predicted* rally intervals against a list of
6:*ground-truth* intervals via temporal IoU and one-to-one greedy matching.
16:     """Temporal intersection-over-union of two intervals.
21:     a_start, a_end = a
22:     b_start, b_end = b
23:     if a_end <= a_start or b_end <= b_start:
25:         inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
28:         union = (a_end - a_start) + (b_end - b_start) - inter
48:def match_intervals(preds: List[Interval], gts: List[Interval],
50:     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
59:         for gi, g in enumerate(gts):
62:             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
64:             candidates.append((iou, pi, gi))
75:             matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
78:         unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
97:         mean_start_error: float     # mean |pred_start - gt_start| over matches, seconds
98:         mean_end_error: float       # mean |pred_end - gt_end| over matches, seconds
104:def score(preds: List[Interval], gts: List[Interval],
112:     mr = match_result if match_result is not None else match_intervals(preds, gts,
iou_threshold)
124:     mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
mr.matches) / tp
125:     mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
mr.matches) / tp
127:     mean_iou = mean_start_err = mean_end_err = 0.0

```

```
131:         num_pred=len(preds), num_gt=len(gts),
134:         mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
```

[Showing results with pagination = limit: 25]

18. assistant (2026-06-16T03:06:46.235Z)

Confirmed. IoU (line 16-28) is computed from differences ``min(a_end,b_end) - max(a_start,b_start)`` and ``mean_start_error`/`mean_end_error`` are ``|pred - gt|`` differences. Every score is **translation-invariant**: adding a constant offset to every ``start`/`end`/`gts`` value in a single fixture leaves all metrics identical. This makes a per-fixture time-origin reset (subtract the fixture's min timestamp so the first event starts at `t=0`, or a random offset) a zero-cost de-linkage of the match timeline. This grounds the "strip absolute timestamps" choice in the actual harness math.

I have all the grounding needed. Submitting the design.

19. user (2026-06-16T03:08:33.666Z)

Structured output provided successfully

20. assistant (2026-06-16T03:08:45.348Z)

The design has been submitted via StructuredOutput.

Key grounding I verified against the actual code (all paths absolute):

- ``C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\fusion_golden.py`` ? ``extract_video`` writes ``name``, ``fps``, ``frame_width``, per-candidate ``shuttle`/`person`/`label``, and ``gts``; ``_derive_video_path`` (lines 39-47) ties ``name`` directly to the source ``.MP4`/`.rallies.csv`` filename ? the primary attribution vector.
- ``C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\hashing.py`` ? ``compute_video_id`` is the MD5 of the whole file (the fingerprinting key the memo flags).
- ``C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\fusion_compare.py:35-52`` ? the harness consumes only ``c['shuttle']``, ``c['start']/c['end']``, ``d['gts']``, and uses ``d['name']`` solely as a ``VideoCandidates`` label, so aliasing ``name`` to an opaque surrogate and dropping the ``person`` block is harness-transparent.
- ``C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\metrics.py:16-28, 124-125`` ? IoU and start/end errors are pure differences, so a per-fixture time-origin reset is provably zero-cost to every regression metric while destroying match-timeline correlatability.
- ``C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\fusion_features.py:194-197`` ? ``PERSON_FEATURE_NAMES`` are aggregate counts/motion (the Art.9 "tripwire" / DPDP singling-out bucket the memo excludes from a public commit), not pose/gait.
- ``C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_DATA_SHARING.md`` and ``eval_baselines/`` confirm the private-features / committed-baselines split the design plugs into.